A Behavioral Comparison of Adversarially Trained ResNet-18 and ViT-Tiny on CIFAR-10: Transfer Metrics, Gradient Structure, and Block-wise Feature Drift

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***Abstract*—How convolutional neural networks (CNNs) and Vision Transformers (ViTs) differ under adversarial attack is usually summarized by a single robust accuracy number. This paper reports preliminary results from an ongoing behavioral comparison of adversarially trained (AT) ResNet-18 and ViT-Tiny on CIFAR-10, evaluated with AutoAttack on the full test set at $\epsilon = 8/255$, and argues that the single number hides most of the story. The first finding is that although the CNN leads in absolute robust accuracy (35.74% vs 32.94%; McNemar $p < 10^{-11}$), a conditional decomposition shows the overall picture to be view-dependent: the entire PGD-10 gap traces to the clean accuracy gap, the per-model conditional ordering under AutoAttack turns slightly in favor of the ViT, yet on the subset both models classify correctly on clean inputs the CNN advantage persists significantly. Repetitions with three independent training runs per architecture confirm the absolute ordering, but show the conditional parity reading to be specific to the checkpoints and to the model selection protocol. The second finding is that cross-architecture transfer asymmetry depends strongly on the conditioning protocol used to measure it: the unconditioned (raw) rate manufactures an artificial 8.3-point asymmetry, target-correct conditioned rates are statistically equivalent, while the both-correct and successful-source protocols reveal a significant 5.3-point CNN→ViT advantage. In short, protocol choice can create, erase, or reverse directional conclusions. The third finding concerns gradient structure: scale-invariant measures show CNN input gradients to be somewhat sparser, while ViT cross-sample gradient alignment remains higher in absolute cosine (the signed cosine is near zero for both models). Finally, under attack ViT features drift with an early drop and mid-network plateau profile, whereas the CNN profile degrades monotonically with depth. Results come from a single training run per architecture and should be read as preliminary; an extended analysis is in preparation.**

***Index Terms*—adversarial examples, adversarial training, CIFAR-10, convolutional neural networks, robustness, transfer attacks, Vision Transformer**

I. Introduction

Deep image classifiers are vulnerable to adversarial examples, small and carefully crafted perturbations that flip predictions with high confidence [1], [2]. As Vision Transformers (ViTs) [3] join convolutional neural networks (CNNs) [4] as a dominant architecture family, a natural question is how the two families differ under attack. Much of the literature approaches this as a contest and reports which architecture attains higher robust accuracy [5]–[7]. This paper takes a different, behavioral view: for a fixed pair of adversarially trained models, *how* do the architectures differ in the composition of the headline robustness number, in how adversarial examples transfer between them, in the structure of their input gradients, and in how perturbations propagate through their layers?

We report preliminary results from an ongoing study of this question on CIFAR-10 [8]: a ResNet-18 and a ViT-Tiny trained with the same adversarial objective and threat model, evaluated with AutoAttack [9] on the full test set. Our contributions are: (i) a conditional decomposition of robustness showing that the "CNN is more robust" headline is composition-dependent; (ii) a multi-protocol analysis showing that the measured direction and size of cross-architecture transfer asymmetry depend on the conditioning protocol, which serves as a methodological caution for transfer studies; (iii) paired, scale-invariant gradient structure comparisons including a signed versus absolute alignment distinction; and (iv) a block-wise feature drift comparison that reveals qualitatively different depth profiles. All results derive from per-sample-logged evaluations; they come from a single training run per architecture and should be read as preliminary.

II. Related Work

Adversarial examples and gradient-based attacks were established by Szegedy et al. [1] and Goodfellow et al. [2]; projected gradient descent (PGD) adversarial training (AT) [10] remains the standard defense, with robust overfitting and early stopping characterized by Rice et al. [11]. AutoAttack [9] and RobustBench [12] standardized robustness evaluation.

Among direct CNN–ViT robustness comparisons, findings vary with the setup: Bai et al. [5] found that under fair training recipes transformers do not surpass CNNs in adversarial robustness, whereas Benz et al. [6] reported ViTs to be more robust than CNNs for standardly trained models. This contrast is consistent with our thesis that comparison outcomes are sensitive to setup and protocol. Closest to our setting, Ali et al. [7] examined the robustness of standardly trained ViTs and CNNs through FGSM (Fast Gradient Sign Method), PGD, and DeepFool score comparisons. Our study differs in three ways: both models are *adversarially trained* under a matched objective; evaluation uses AutoAttack on the full test set with paired per-sample statistics; and the comparison goes beyond scores to conditional decompositions, protocol-explicit transfer, gradient structure, and block-wise drift. ViT adversarial training is known to be recipe-sensitive on small datasets [13], [14], and with synthetic extra data generated by diffusion models ViTs can surpass WideResNets on CIFAR-10 [15]. Our conclusions therefore hold under the baseline recipe we study, not for optimally trained ViTs.

For transfer evaluation, conditioning on correctly classified samples is established practice [16]–[19]; both-correct variants were used by Mahmood et al. [18] and TREND [19]. Our contribution is to show on a single dataset that the *choice* among these protocols qualitatively changes the asymmetry conclusion.

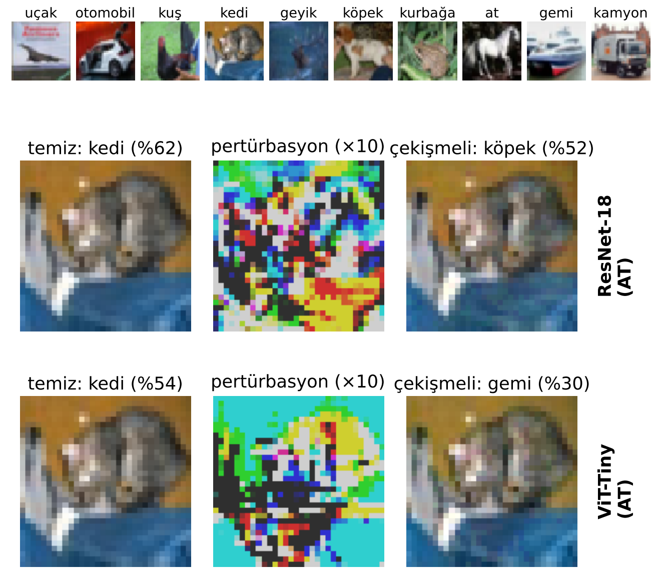

Fig. 1. Top: one sample from each CIFAR-10 class. Bottom: each model's own white-box PGD-10 attack ($\epsilon = 8/255$) on the same test image, with the clean prediction, the perturbation ($10\times$), and the adversarial prediction.

III. Method

A. Dataset, Models, and Training

Experiments are conducted on CIFAR-10 [8]: 60,000 color images of size 32×32 in ten balanced classes (50,000 train / 10,000 test). Fig. 1 illustrates the dataset and the effect of the attack. The example shown is the first sample in test order that is correctly classified on clean input and whose class the attack flips, a deterministic selection criterion; the perturbations are hard to notice at native scale yet change predictions with high confidence, and they are qualitatively different in structure across the two architectures (diffuse for the CNN, patch-aligned for the ViT). Images are 32×32 and are displayed with pixel-preserving (nearest neighbor) magnification. We compare a ResNet-18 (11.2M parameters; CIFAR-style stem) with a ViT-Tiny (5.7M parameters; `timm` [20] architecture, 16×16 patches over inputs bilinearly upsampled to 224×224); both models are trained from scratch on CIFAR-10 with basic data augmentation and then adversarially fine-tuned. All operations including attacks are defined in the native 32×32 CIFAR-10 pixel space: the $32 \to 224$ bilinear upsampling is a differentiable layer *inside* the ViT wrapper (`nn.Upsample`). The two models therefore share the same L_∞ threat model ($\epsilon = 8/255$ in raw $[0, 1]$ pixel space), and transferred adversarial examples are fed to both models as the same 32×32 inputs.

Adversarial training minimizes the mixed objective $(1 - \lambda)\mathcal{L}_{CE}(f(x), y) + \lambda \mathcal{L}_{CE}(f(x_{adv}), y)$ with $\lambda = 0.5$, where x_{adv} is generated in training mode with PGD-10 ($\epsilon = 8/255$, $\alpha = 2/255$, random start from $\mathcal{U}(-\epsilon, \epsilon)$, single restart) following standard practice [10], [11]. Clean pretraining runs for 200 epochs (CNN: SGD, learning rate 0.1, weight decay 5×10^{-4}; ViT: AdamW [21], 10^{-3}, decay 0.05; cosine annealing); adversarial fine-tuning runs for at most 100 epochs with learning rate 10^{-3} and batch size 128 (CNN) / 64 (ViT). Data augmentation is random cropping with 4-pixel padding and horizontal flipping. Early stopping and model selection are driven by PGD-10 accuracy on a 2,000-sample validation split held out from the training set (patience 20, threshold 0.1 points); the test set is touched only for final evaluations. As an upper reference we include the WideResNet-28-10 checkpoint of Gowal et al. [22] distributed through RobustBench [12]; since that model is trained with extra data and a stronger recipe, it is kept visually and analytically separate from the matched pair.

B. Evaluation and Metrics

All evaluations are run on the full 10,000-sample test set with per-sample outcome logging. Robustness is measured at $\epsilon = 8/255$ with FGSM, PGD-10, and the standard AutoAttack (AA) ensemble (official implementation, version 0.1: APGD-CE, APGD-T, FAB-T, Square). Alongside absolute robust accuracy we report the *conditional fooling rate*, that is, how many of the samples classified correctly on clean input the attack overturns. Since samples misclassified on clean input count as non-robust, absolute robustness factors exactly as robust $=$ clean $\times$ $(1 -$ conditional fooling$)$, separating the clean accuracy component from the conditional sensitivity component. We further evaluate the *both-correct* subset, the samples both models classify correctly on clean input, which enables one-to-one paired comparisons.

For transfer, adversarial examples generated on a source model are evaluated on the other (target) model under three conditioning protocols: *target-correct* (the target classifies the clean sample correctly), *both-correct* [18], [19], and *successful-source* (the target is correct and the attack succeeds

TABLE I
ROBUSTNESS ON THE FULL CIFAR-10 TEST SET ($\epsilon = 8/255$).

Model	AT	Clean	FGSM	PGD-10	AA
ResNet-18	–	94.37	30.93	0.01	–
ViT-Tiny	–	77.50	3.34	0.03	–
ResNet-18	✓	85.42	49.90	40.97	**35.74**
ViT-Tiny	✓	75.65	40.07	36.09	**32.94**
WRN-28-10†	✓	89.48	70.91	66.92	62.76

†External reference checkpoint [22]: stronger recipe and extra data; the AA value is as reported by RobustBench.

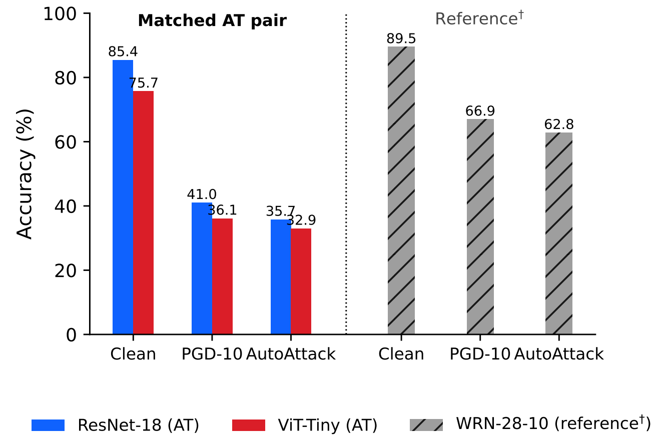

Fig. 2. Clean, PGD-10, and AutoAttack accuracies. WRN-28-10 (hatched) is an external reference trained with a stronger recipe and extra data.

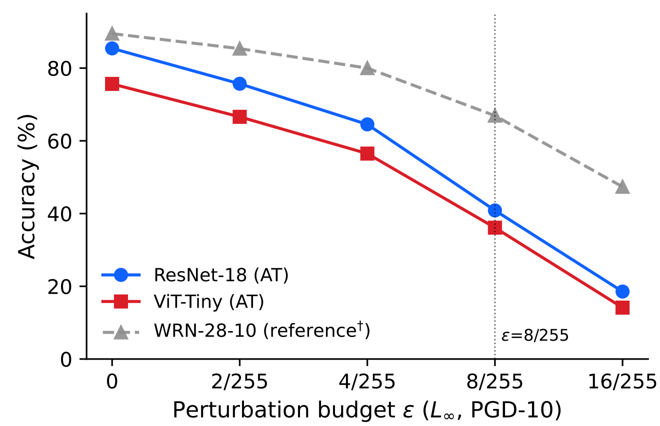

Fig. 3. PGD-10 accuracy as a function of the perturbation budget (full test set).

in the white-box sense on the source). Confidence intervals (CIs) are given by percentile bootstrap (10,000 resamples). Under the both-correct protocol the fooling indicators of the two directions are defined on the *same* samples: the paired bootstrap resamples the members of the joint set, and the sign-flip permutation test is built on per-sample directional differences. The main ± 2 point margin of the two one-sided equivalence tests (TOST) was chosen on the principle of treating as practically equivalent any difference below the variation observed at the training-run level (a few points); the ± 1 and ± 3 sensitivities are also reported in the text.

Gradient structure is examined through per-sample loss gradients on 500 fixed test samples: scale-invariant sparsity (Hoyer, Gini [23], and the fraction of components below 1% of the per-sample maximum) with per-sample paired Wilcoxon tests (Holm-corrected); cross-sample gradient alignment is reported as the mean absolute cosine over all sample pairs, together with the signed mean. Block-wise feature drift is measured on the first 100 test samples (unselected, regardless of correctness) using each model's *own* white-box PGD-10 examples: for the ViT, the MLP sub-block output of each transformer block is flattened per sample over all tokens (CLS + 196 patches); for the CNN, each residual block output is flattened over the spatial dimensions. We then compute the cosine similarity and the norm change between clean and adversarial representations (12 blocks for the ViT, 8 residual blocks for the CNN).

IV. RESULTS

A. Robustness: One Headline, Three Readings

Table I and Fig. 2 give the absolute results: the adversarially trained CNN leads the ViT under every attack (AutoAttack 35.74% vs 32.94%; on joint samples McNemar $\chi^2 = 48.65$, $p < 10^{-11}$), and both trail the reference WideResNet, whose advantage is the joint product of capacity, extra data, and a stronger recipe [22]. Fig. 3 shows the gap to be stable across perturbation budgets.

The conditional decomposition (Table II) changes the reading. Under PGD-10 the conditional fooling rates of the two models are statistically indistinguishable (52.15% vs 52.33%), so the entire absolute PGD gap is explained by clean accuracy (85.42×0.478 vs 75.65×0.477). Under AutoAttack the per-model conditional ordering even turns slightly in favor of the ViT (fooling 58.16% vs 56.46%). Yet on the 7,260 jointly solved samples, which provide a one-to-one paired comparison, the CNN leads significantly under both attacks (PGD 54.92% vs 49.61%, McNemar $p < 10^{-20}$; AA 48.15% vs 45.33%, $p < 10^{-6}$). These views reconcile compositionally: the ~1,300 samples only the CNN solves on clean input are almost never robust (PGD survival below 8%) and pull down the CNN per-model conditional rate. A single robust accuracy number would hide all of this structure.

To test how stable this conditional reading is, we repeated the training pipeline with three independent runs per architecture after the analyses above were completed; in these repetitions the validation split was held out before the clean pre-training of every model, which removes a leak in model selection. The absolute ordering is preserved (AutoAttack 37.9±0.1% vs 29.1±0.4%), but the conditional parity is not: under the corrected protocol the conditional fooling rate is *lower* for the CNN (PGD 48.6±0.8% vs 55.5±0.5%; AA 55.8±0.2% vs 60.4±0.3%), and the both-correct gap widens rather than shrinks. The reading that conditional sensitivity is matched is therefore specific to the checkpoints and to

TABLE II
CONDITIONAL DECOMPOSITION (FULL TEST; 95% BOOTSTRAP CIS). BOTH-CORRECT: THE $n = 7{,}260$ SAMPLES BOTH MODELS CLASSIFY CORRECTLY ON CLEAN INPUT.

Model	Conditional fooling (%)		Both-correct robust (%)	
	PGD-10	AA	PGD-10	AA
ResNet-18 AT	52.15 [51.1; 53.2]	58.16 [57.1; 59.2]	**54.92**	**48.15**
ViT-Tiny AT	52.33 [51.2; 53.5]	**56.46** [55.4; 57.6]	49.61	45.33

TABLE III
TRANSFER ASYMMETRY UNDER THE UNCONDITIONED RATE AND THREE CONDITIONING PROTOCOLS (PGD-10, $\epsilon = 8/255$, FULL TEST; 95% BOOTSTRAP CIS).

Protocol	CNN→ViT	ViT→CNN	Diff.
Unconditioned (raw)	40.17	31.90	+8.27
Target-correct	20.95 [20.03; 21.88]	20.32 [19.47; 21.19]	+0.63
Both-correct	19.19 [18.26; 20.10]	13.86 [13.07; 14.66]	+5.33
Successful-source	39.07 [37.47; 40.66]	33.79 [32.48; 35.13]	+5.28

Denominators (CNN→ViT / ViT→CNN): unconditioned $N = 10{,}000/10{,}000$; target-correct $N = 7{,}565/8{,}542$; both-correct $N = 7{,}260$ (joint set); successful-source $N = 3{,}576/4{,}936$.

the selection protocol; what remains robust is that the single-number comparison depends on which view is reported. This also illustrates the paper's thesis from a second angle: not only the evaluation protocol but the *training* protocol can reverse a directional conclusion. The full seed analysis is deferred to the extended version.

B. Transfer: The Protocol Is Part of the Result

Table III reports cross-architecture transfer under three established conditioning protocols alongside the unconditioned (raw) rate, with the denominator of each protocol given in the table footnote. The unconditioned (raw) rates suggest an 8.3-point asymmetry in favor of CNN sources (40.2% vs 31.9%), an artifact produced by the ten-point clean accuracy gap between the targets. The target-correct protocol removes this confounder and shows near parity (difference 0.63 points; TOST equivalence at a ± 2 point margin, $p = 0.016$); this parity is also stable under the momentum-based MI-FGSM [17] (19.85% vs 20.11%). The both-correct and successful-source protocols, however, *turn the difference back in favor of the CNN on the joint set*: on jointly solved samples the directions separate markedly (∼5.3-point CNN→ViT advantage; paired bootstrap CI [4.39; 6.25]; sign-flip permutation $p < 5 \times 10^{-5}$; TOST equivalence is rejected even at ± 3 points). The mechanism is compositional: samples the target solves on clean input but the source does not are extremely fragile in both directions (56.9% and 63.0% fooling), and the asymmetric sizes of these marginal subsets (1,282 vs 305) pull the target-correct rates toward parity. The practical lesson for cross-architecture transfer studies is that the conditioning protocol can create, erase, or expose a directional difference on the same data; it should therefore be stated explicitly and, where possible, varied.

C. Gradient Structure

On the 500 fixed test samples measured for both models, CNN input gradients are consistently but moderately sparser than those of the ViT under all three scale-invariant measures (Hoyer 0.474 ± 0.009 vs 0.449 ± 0.011; Gini 0.634 vs 0.606; components below 1% of the per-sample maximum 31.5% vs 25.8%), and the difference is significant under paired Wilcoxon tests (Holm-corrected $p < 10^{-15}$; paired Cohen's d of 0.34–0.65). Adversarial training is what creates this ordering: in the cleanly trained pair the sparser model is the ViT (Hoyer 0.406 vs 0.354), which is consistent with the strong sparsifying effect of adversarial training on CNN gradients [24]. Cross-sample gradient alignment, as mean absolute cosine over all pairs, is higher for the ViT (0.052 vs 0.038); observing nearly identical levels in the cleanly trained pair indicates that for the checkpoints examined this difference does not stem from adversarial training alone. The *signed* mean cosine, however, is near zero for both models (≤ 0.002): what the samples share is axes of sensitivity, not consistent directions.

D. Block-wise Feature Drift

Fig. 4 compares how far the representation of an attacked input departs from its clean counterpart at each block. The ViT shows an early drop and mid-network plateau profile: cosine similarity falls rapidly to a minimum of 0.955 at block 8, then recovers slightly (0.965 at blocks 10–11); feature norms change by less than 1.5% throughout, so the perturbation corrupts the *direction* of the representation rather than its magnitude. The CNN behaves qualitatively differently: cosine similarity falls monotonically and with acceleration through depth (from 0.995 to 0.913 at the last residual block), and large norm distortions appear in the final two blocks (-6.4% and $+7.2\%$). This plateau and recovery pattern is not a general property of deep networks; it is specific to the ViT in the model pair examined. The analysis carries layer-wise representation studies of cleanly trained ViTs [25] over to the adversarially trained setting. The shaded bands in Fig. 4 show ± 1 standard deviation over the mean of 5 batches.

V. DISCUSSION AND LIMITATIONS

Three main implications stand out. First, single-number robustness comparisons are lossy: on the checkpoints studied here, our conditional decomposition simultaneously supports all three readings of "the CNN is more robust" (absolute and both-correct views), "conditional sensitivity is matched" (per-model PGD), and "the ViT loses less on the samples it solves" (per-model AutoAttack); that the latter two readings disappear in the three-seed repetitions teaches the same lesson from the opposite direction, namely that which view is reported is part of the result. In addition, claims of transfer asymmetry are relative to the protocol; cross-architecture studies should report the conditioning protocol as part of the result. The final implication is a behavioral dissociation: the architectures differ in gradient concentration and in where adversarial damage

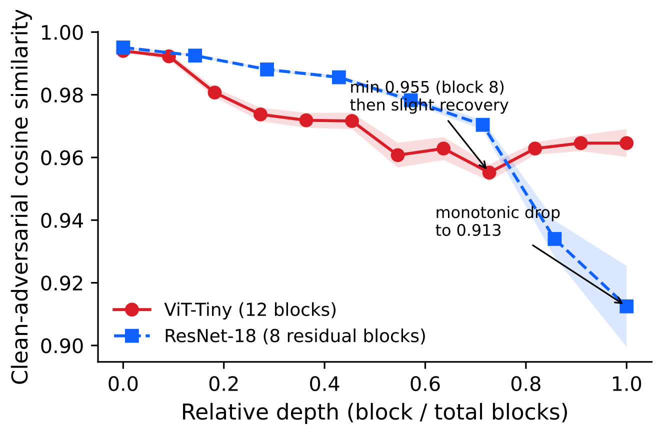

Fig. 4. Block-wise cosine similarity between clean and adversarial representations as a function of relative depth ($n = 100$; bands are ± 1 standard deviation).

accumulates along depth, which suggests stabilization targeted at different locations for the two architectures.

Several limitations frame these results. The findings rest on a single training run per architecture under a single baseline recipe, and are therefore indicative rather than settled. Moreover, all statistical tests used (McNemar, Wilcoxon, bootstrap, permutation) infer at the sample level and are *conditional on the fixed trained checkpoints*; they do not quantify training seed uncertainty. ViT robustness is also known to be recipe-sensitive [13]–[15]. The study covers a single dataset, a single model pair, and a single-budget L_∞ threat model for the conditional analyses. The gradient and drift measurements are behavioral correlations, not established causal explanations. Our first replications with three independent training runs per architecture confirm the absolute ordering but show the conditional parity reading to be checkpoint-dependent (Section IV-A); the transfer, gradient, and drift analyses of those repetitions, together with an extended analysis addressing additional transfer attacks and spatial gradient statistics, are in preparation.

VI. Conclusion

For a matched adversarially trained ResNet-18 / ViT-Tiny pair on CIFAR-10 we showed that the robustness headline is composition-dependent, that the measured transfer asymmetry is largely a property of the conditioning protocol, that gradient structure differences are consistent but moderate (with the alignment difference remaining unsigned by nature), and that adversarial damage accumulates at qualitatively different depths in the two architectures. We offer these preliminary results as a template for protocol-explicit, multi-view robustness comparisons across architecture families.

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